Forces and Inclines

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1 Review of Spring Force

1. The spring force is $\vec{s} = -k\Delta\vec{x}$. That is, the force is directly proportional to the *change* in length. Use the given weights, spring, ruler and hook to measure the spring constant k for your spring. That is, after drawing an appropriate free-body diagram, we find that

$$|mg| = k|\Delta x| \tag{1}$$

Solving for k:

$$k = \frac{mg}{\Delta x} \tag{2}$$

2. The constant k has units of Newtons per meter. Measure k and record the value here, accounting for errors:

2 Inclined Surfaces

1. Place your weights on the ruler and align the spring with the ruler such that the ruler serves as an *inclined plane* for the weights.
2. Use enough weight so that you can observe increases in the length of the spring, even when the incline angle is small.
3. Draw a free-body diagram for the weight:

3 Net Force on Weight

1. The net force *down the ruler* should be $F_{net} = mg\sin\theta$, where θ is the angle between the ruler and the table. Use the protractor to measure the angle between the ruler and the table.

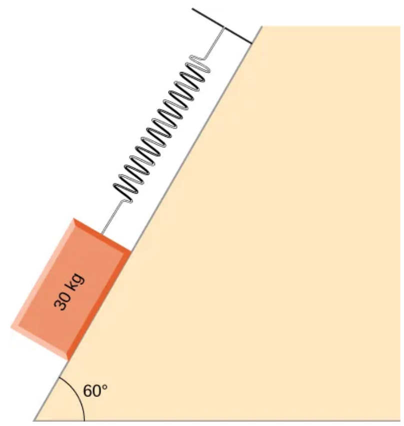

Figure 1: A spring holding a mass on an incline.

2. Create a graph below of the Δx (change in length) of the spring versus θ. Does it follow the expected $\sin\theta$ dependence?

4 Modification to the Experiment

1. What effect is missing in the experimental analysis? How would you measure account for it in your graph?